

Name: Aryan More Class: Soc 9 Batch: A
Subject: Programming Lang II Roll No.: 11 Exp. No.:
Name of the Experiment: Assignment - 02
Performed on: _____
Submitted on: _____

Marks	Teacher's Signature with Date
<u>10</u>	<u>[Signature]</u>

1. List the python sequence data type and state one key characteristic of each with respect to mutability or ordering?

=> i) List :- Ordered and mutable (elements can be changed)

ii) Tuple :- Ordered and immutable (cannot be modified)

iii) String :- Ordered and immutable sequence of characters

iv) Set :- Unordered, mutable, store unique elements only.

v) Dictionary :- Ordered (Python 3.7+), mutable, stores data as key value pairs.

2. Explain how indexing and slicing work in python sequences. Illustrate how they

Incomplete for Theory ☐ Diagrams ☐ Observation Table ☐

Calculations ☐ Graphs ☐ Results ☐ Conclusion ☐

Understanding Through Q/A ☐ Late Submission ☐ Neatness ☐

[Signature]
Teacher's Signature

Indexing :

`s[1] → 'y'`

`l[2] → 30`

Slicing :

`s[1:4] → "y#"`

`l[1:3] → [20, 30]`

Difference :-

- string cannot be modified after slicing (immutable)
- List can be modified after slicing (mutable).

differ when applied to string and list?

=> Indexing:- It is used to access a single element using its position.
(start from 0).

Slicing:- It is used to access a part of the sequence using [start: end: step]

Example:-

S = "Python"

I = [10, 20, 30, 40]

3. Write a python program that:
- Uses of indexing and slicing on a string list, and
 - Applies atleast two built-in function to apply display meaningful results.

=> ^{Aryan}
text = "Samarth"
print ("First character:", text[0])
print ("Substring:", text[0:3])
print ("Length of string:", len(text))
print ("Uppercase:", text.upper())

output:

First character: S

Substring: Ary

Length of string: 7

Uppercase: ARYAN

4. Analyze the difference between equality ($==$) and identity (is) when used with sequence type. Discuss how object mutability and memory reference affect program behaviour?

=> Equality ($==$) checks whether values are same.

Identity (is) checks whether both variable point to the same memory location,

Example -

`a = [1, 2, 3]`

`b = [1, 2, 3]`

`print(a == b)`

`print(a is b)`

Output:

True

False

Explanation:-

- Lists are mutable and stored in different memory location.

- $==$ compares content
- is compares memory reference.

5. Evaluate the suitability of lists, sets, and dictionaries for storing and managing unique student records.

=> List: Allows duplicates, slower searching
Set: Stores only unique value, no indexing
Dictionaries: Stores unique student IDs as key and detail as value.



6. Design and implement a python program that uses:
- Appropriate sequence type
 - Indexing or slicing.
 - Built-in function to manage and process student data efficiently.

```
Students = { 101: "Amit", 102: "Neha", 103: "Rohul" }
```

```
print (" Student 101: ", students[101])  
print (" Total students: ", len (students))  
print (" Student IDs: ", list (students. key (1))
```

Output:

Student 101: Amit

Total students : 3

Student IDs : [101, 102, 103]

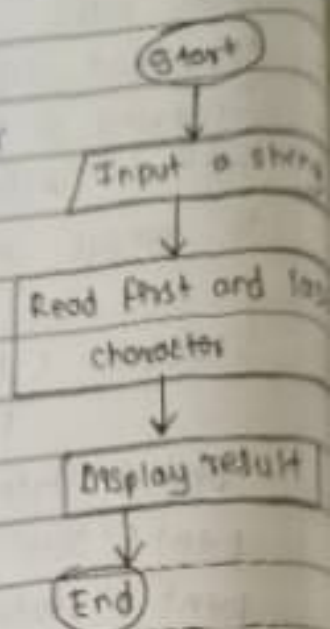
Write a program to print the first and last character of a string?

```
S = input (" Enter a string: ")  
print (" First character: ", S[0])  
print (" Last character: ", S[-1])
```

Output : First character : p
Last character : n

→ Algorithm:-

- 1) Start
- 2) Read a string
- 3) Find the first character
- 4) Find the last character
- 5) Display result
- 6) Stop



Q. write a program to create a list, tuple, set and dictionary?

→ list = [1, 2, 3]

tuple = (4, 5, 6)

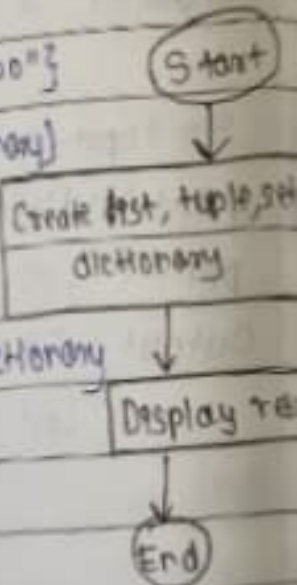
string = "1, 2, 3"

dictionary = {1: "one", 2: "two"}

print (list, tuple, string, dictionary)

→ Algorithm:-

- 1) Start
- 2) Create a list, tuple, set, dictionary
- 3) Display result
- 4) Stop



→ Output: [1, 2, 3]

(4, 5, 6)



Design and implement a python program that uses:

Appropriate sequence type

Indexing or slicing

Built-in function

to manage and process student data efficiently.

```
Students = { 101: "Amit", 102: "Neha", 103: "Rohul" }
```

```
print ("Student 101: ", students[101])
print ("Total students:", len(students))
print ("Student IDs:", list(students.keys()))
```

Output:

Student 101: Amit

Total students : 3

Student IDs : [101, 102, 103]

Write a program to print the first and last character of a string?

```
S = input ("Enter a string:")
print ("First character:", S[0])
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Output: First character: p

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$a = [1, 2, 3]$

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$print(a == b)$

$print(a is b)$

Output:

True

False

Explanation:-

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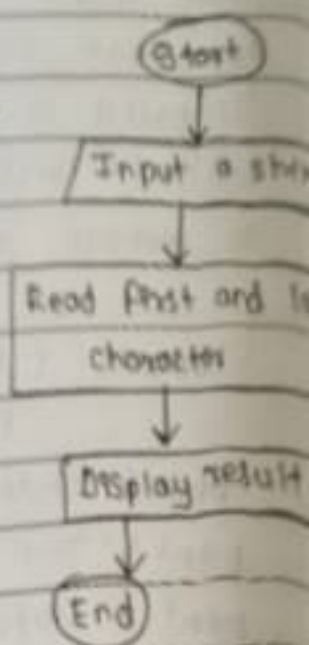
- ($==$) compares content
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5. Evaluate the suitability of lists, sets, and dictionaries for storing and managing unique student records.

=> List: Allows duplicates, slower searching
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Algorithm:-

- 1) Start
- 2) Read a string
- 3) Find the first character
- 4) Find the last character
- 5) Display result
- 6) Stop



Q. write a program to create a list, tuple, set and dictionary?

list = [1, 2, 3]

tuple = (4, 5, 6)

string = [7, 8, 9]

dictionary = {1: "one", 2: "two"}

print (list, tuple, string, dictionary)

Algorithm:-

- 1) Start
- 2) Create a list, tuple, set, dictionary
- 3) Display result
- 4) Stop

Create list, tuple, set, dictionary

Display result

End

Output: [1, 2, 3]

(4, 5, 6)



List the key features and advantages of Python. Also mention any two Python IDEs or editors used for program development.

Features of Python are:-

- Simple and easy to learn syntax.
- Interpreted and high-level language.
- Object-oriented and modular.
- Open source and free.

Advantages of Python are:-

- Faster development time.
- Easy debugging and maintenance.
- Strong community support.
- Used in web development, AI, ML, data science, automation.

Python IDEs / Editors

Python Pycharm

IDE

Explain the history and evolution of Python. Describe why Python has become popular for beginner and industry application.

Python was developed by Guido van Rossum in the late 1980s and early 1990s. It was designed to emphasize code readability and simplicity.

- Evolution:
- Python 1.x - Basic feature
 - Python 2.x - Enhanced performance, libraries.
 - Python 3.x - Better syntax, Unicode support, improved security.

Why Python is popular:

- Easy for beginners.
- English like syntax.
- widely used in industry.
- supports multiple programming paradigms.

Q.3) Write a python program to accept numeric values from the user and perform the following operations: Addition, Multiplication, Division. Display the results using proper input and output.

- ```
a = float(input("Enter first number: "))
b = float(input("Enter second number: "))
print("Addition: ", a + b)
print("Subtraction: ", a - b)
print("Multiply: ", a * b)
print("Division: ", a / b)
```

Analyze the following python code and identify the data types of each variable. Also explain the output.

1.  $a = 10$
2.  $b = 5.5$
3.  $c = 3 + 4j$
4.  $d = \text{True}$
5. `print (a+b, type (c), d)`

Data types:

$a \rightarrow \text{int}$

$b \rightarrow \text{float}$

$c \rightarrow \text{complex}$

$d \rightarrow \text{bool}$

Output Explanation:

$a+b \rightarrow 15.5$

`type (c) \rightarrow <class 'complex'>`

$d \rightarrow \text{True}$

Output: Python

`15.5 <class 'complex'> True`

Evaluate the use of relational, logical and bitwise operators in python. Justify with suitable examples with each type of operator is most appropriate.



List the key features and advantages of Python. Also mention any two Python IDEs or editors used for program development.

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Object-oriented and modular.  
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## History of Computing

- Python was developed by Guido van Rossum in the late 1980s and early 1990s.
- It was designed to be simple and easy to learn.
- Python 2.x - Basic feature
- Python 3.x - Enhanced performance and libraries.
- Python 3.x - Better syntax, Unicode, Super, improved security

Why Python is popular:

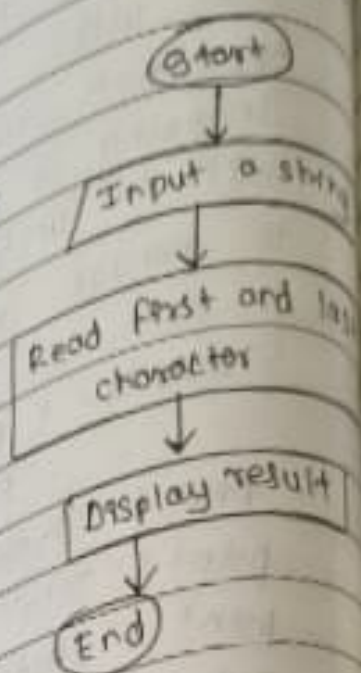
- Easy to learn.
- Easy to use.
- Used in industry.
- Support multiple programming paradigms.

Ex: Write a Python program to accept two numbers from the user and perform the following operations: Addition, Subtraction, Division, Display the result using proper input and output prompts.

```
a = int(input("Enter first number: "))
b = int(input("Enter second number: "))
print("Addition: ", a + b)
print("Subtraction: ", a - b)
print("Multiplication: ", a * b)
print("Division: ", a / b)
```

### Algorithm:-

- 1) Start
- 2) Read a string
- 3) Find the first character
- 4) Find the last character
- 5) Display result
- 6) Stop



Q. write a program to create a list, tuple, set and dictionary?

→ list = [1, 2, 3]  
 tuple = (4, 5, 6)  
 string = {7, 8, 9}  
 dictionary = {1: "one", 2: "two"}  
 print (list, tuple, string, dictionary)

### Algorithm:-

- 1) Start
- 2) Create a list, tuple, set, dictionary
- 3) Display result
- 4) Stop

→ Output: [1, 2, 3]  
 (4, 5, 6)

